Strictness of the Hartman–WAP Inclusion: A Complete LCA Classification

Candidate counterexample/full classification fa_banach_001, lane 8

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Source problem. Maresch and Winkler, arXiv:math/0510064, Problem 5.4.7, ask when

$$\mathcal{H}_c(G) \supseteq \mathcal{H}(G) \cap \mathcal{W}(G)$$

is strict and request a function in the difference. Their Definition 5.4.2 says that $\mathcal{H}_0(G)$ consists equivalently of Hartman functions admitting a Riemann-integrable realization supported on a meager Haar-null set; Lemma 5.4.4 proves $\mathcal{H}_0(G) \subseteq \mathcal{H}_c(G)$.

2. G is minimally almost periodic.

Proof. For map G the Bohr compactification $bG = \{0\}$ consists of only one element. ☐

PROBLEM 5.4.7. *For which topological groups is the inclusion $\mathcal{H}_c(G) \supseteq \mathcal{H}(G) \cap \mathcal{W}(G)$ strict? Construct $f \in \mathcal{H}_c(G) \setminus (\mathcal{H}(G) \cap \mathcal{W}(G))$.*

LEMMA 5.4.8. *Let G be a non compact topological group and let (ι_b, bG) be the Bohr compactification of G.*

Problem 5.4.7, source PDF page 54.

Theorem 1. *Let G be an infinite abelian topological group whose continuous characters separate points. Then there exists a 0–1 function*

$$f \in \mathcal{H}_0(G) \subseteq \mathcal{H}_c(G), \qquad f \notin \mathcal{W}(G).$$

Consequently, for a locally compact abelian group G, the inclusion in Problem 5.4.7 is strict if and only if G is infinite.

Lemma 1. *If Γ_0 is a finite set of continuous characters of an infinite abelian group G, then every set*

$$U = \{g \in G : |\chi(g) - 1| < \varepsilon \text{ for all } \chi \in \Gamma_0\}, \qquad \varepsilon > 0,$$

is infinite.

Proof. Let $q : G \to \mathbb{T}^{\Gamma_0}$ be the character map. If $q(G)$ is finite, then $\ker q$ is infinite and is contained in U. If $q(G)$ is infinite, its identity is non-isolated: otherwise the precompact group $q(G)$ would be discrete, hence finite. Thus the indicated identity neighborhood meets $q(G)$ infinitely often, and so does its preimage. ☐

Proof of Theorem 1. We first construct points $x_n, y_n \in G$ and finite character sets Γ_n. Start with $\Gamma_0 = \varnothing$. Suppose the construction is complete through stage $n-1$, and put

$$U_n = \{g \in G : |\chi(g) - 1| < 1/n \text{ for every } \chi \in \Gamma_{n-1}\}.$$

By Lemma 1, U_n is infinite. Choose $x_n, y_n \in U_n$ so that all sums

$$x_i + y_j, \qquad 1 \le i, j \le n,$$

are distinct. This is possible because, once the old points are fixed, every collision involving x_n or y_n excludes only one of finitely many values; also avoid the finitely many old x_i and y_j. More explicitly, first choose x_n outside the finite set that would make one of the sums $x_n + y_j$ ($j < n$) collide with an old sum, and outside the old x_i's. Then choose y_n outside the finite set that would make one of the sums $x_i + y_n$ ($i \le n$) collide with an old sum or with one of the new sums $x_n + y_j$ ($j < n$), and outside the old y_j's.

For every nonzero difference between two of these finitely many sums, choose a continuous character which is nontrivial on that difference, and let Γ_n be Γ_{n-1} together with all such characters. This uses exactly the hypothesis that continuous characters separate points.

Let $\Gamma = \bigcup_n \Gamma_n$, define

$$\iota : G \longrightarrow \mathbb{T}^\Gamma, \qquad \iota(g) = (\chi(g))_{\chi \in \Gamma},$$

and set $C = \overline{\iota(G)}$. Since Γ is countable, C is a metrizable compact group and (ι, C) is a group compactification. Every two selected sums $x_i + y_j$ have distinct images: their difference was separated at some finite stage. Moreover

$$\iota(x_n) \longrightarrow e_C, \qquad \iota(y_n) \longrightarrow e_C.$$

Indeed, each fixed $\chi \in \Gamma$ belongs to Γ_N for some N, and its error on x_n, y_n is less than $1/n$ for all $n > N$.

Define the upper-triangular support

$$S = \{\iota(x_n + y_m) : n \le m\} \subset C, \qquad F = \mathbf{1}_S, \qquad f = F \circ \iota.$$

We claim that

$$\overline{S} \subseteq S \cup \{\iota(x_n) : n \in \mathbb{N}\} \cup \{e_C\}. \tag{1}$$

Because C is metrizable, it suffices to inspect convergent sequences from S. After passing to a subsequence, either the pairs (n_k, m_k) are constant, giving a point of S; or $n_k = n$ is constant while $m_k \to \infty$, giving $\iota(x_n)$; or $n_k \to \infty$, hence also $m_k \to \infty$, giving e_C. This proves (1).

The group C is infinite because it contains infinitely many distinct selected sums. Hence it has no isolated points and its Haar measure has no atoms. The right-hand side of (1) is countable, therefore meager and Haar-null. The discontinuity set of $F = \mathbf{1}_S$ is contained in $\overline{S}$, so F is Riemann integrable; moreover $[F \neq 0] = S$ is meager and Haar-null. By the source's existential characterization,

$$f \in \mathcal{H}_0(G) \subseteq \mathcal{H}_c(G).$$

Separation of all selected sums gives the exact matrix

$$f(x_n + y_m) = \begin{cases} 1, & n \le m, \\ 0, & n > m. \end{cases}$$

Therefore both iterated limits exist, but

$$\lim_{n\to\infty} \lim_{m\to\infty} f(x_n + y_m) = 1, \qquad \lim_{m\to\infty} \lim_{n\to\infty} f(x_n + y_m) = 0.$$

Grothendieck's double-limit criterion, Proposition 3.6.5 of the source, yields $f \notin \mathcal{W}(G)$.

Every LCA group has separating continuous characters by Pontryagin duality, proving strictness for every infinite LCA group. If G is finite, every bounded function is continuous and almost periodic, so

$$\mathcal{H}_c(G) = \mathcal{H}(G) \cap \mathcal{W}(G) = B(G).$$

This proves the classification. $\square$

Concrete instance on $\mathbb{Z}$. Fix irrational α. Choose positive return times r_j with

$$r_j > 2 \sum_{i<j} r_i, \qquad \|\alpha r_j\|_{\mathbb{R}/\mathbb{Z}} < 1/j,$$

and set $x_n = r_{2n-1}$, $y_n = r_{2n}$. Rapid growth makes all two-term sums unique, while irrational rotation sends both sequences to the identity. The preceding proof then applies with $\iota(k) = e^{2\pi i \alpha k}$.

Verification and novelty. The adversarial report checks every proof step and gives verdict *valid* (97/100). The script `code/verify_z_instance.py` passes 120 exact two-sum checks and an 8×8 translate-matrix test. Exact-phrase, title/citation, and topic searches found the source and surveys but no later answer to Problem 5.4.7 or this LCA classification. This is a bounded novelty check, not an exhaustive priority claim.

References

[1] G. Maresch and R. Winkler, *Compactifications, Hartman functions and (weak) almost periodicity*, arXiv:math/0510064; Dissertationes Math. 461 (2009), 1–72.

[2] A. Grothendieck, *Critères de compacité dans les espaces fonctionnels généraux*, Amer. J. Math. 74 (1952), 168–186.